Technische Universität Berlin
Fakultät VI – Planen Bauen Umwelt
Institut für Landschaftsarchitektur und Umweltplanung
Fachgebiet KI und Landnutzungswandel

Wissenschaftliche Arbeit zur Erlangung des Grades
Master of Science (M.Sc.)

Daily Extreme Precipitation Events and Long-Term Climate Trends across the Iberian Peninsula and Greece

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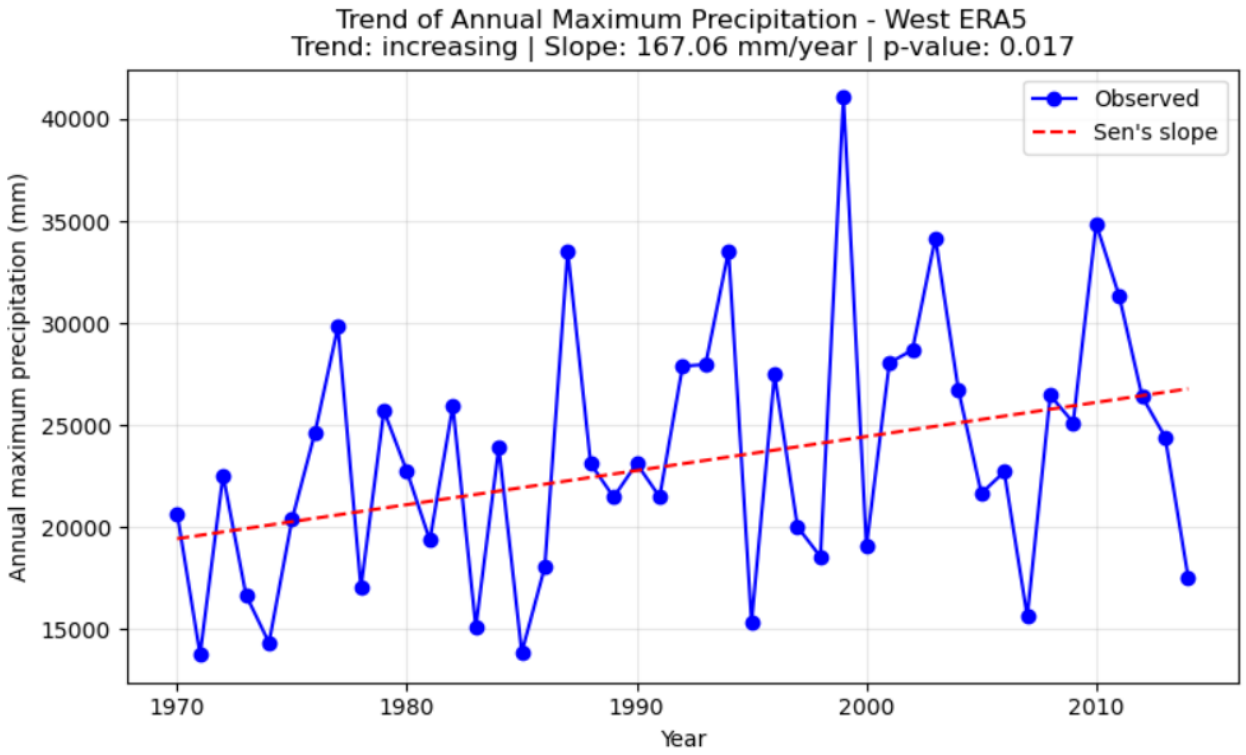
1. Are there observable **trends** in the frequency of daily precipitation extremes in each region over time?

Mann Kendell Test + Sen slope

trend = increasing / decreasing / no trend
p < 0.05 → statistically significant trend
slope → mm/year (or precipitation units)

	trend	p value	tau	slope per year	intercept
daily sums west	increasing	0.017449	0.246465	167.062778	19426.134524
daily sums east	no trend	0.323146	0.10303	79.085109	16415.788144



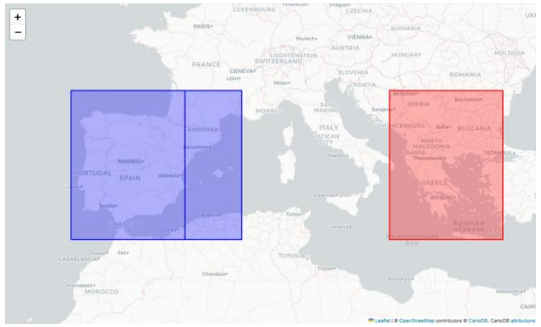
3. How well does the type-I generalized extreme value distribution (= gumbel distribution) represent extreme rainfall days in the Iberian Peninsula and Greece, and what **return periods** can be derived from it? Gumbel and return periods

$$F(x) = e^{-e^{-y}}$$

$$y = \frac{x - \mu}{\beta}$$

scipy.stats.gumbel_r

A right-skewed Gumbel continuous random variable.



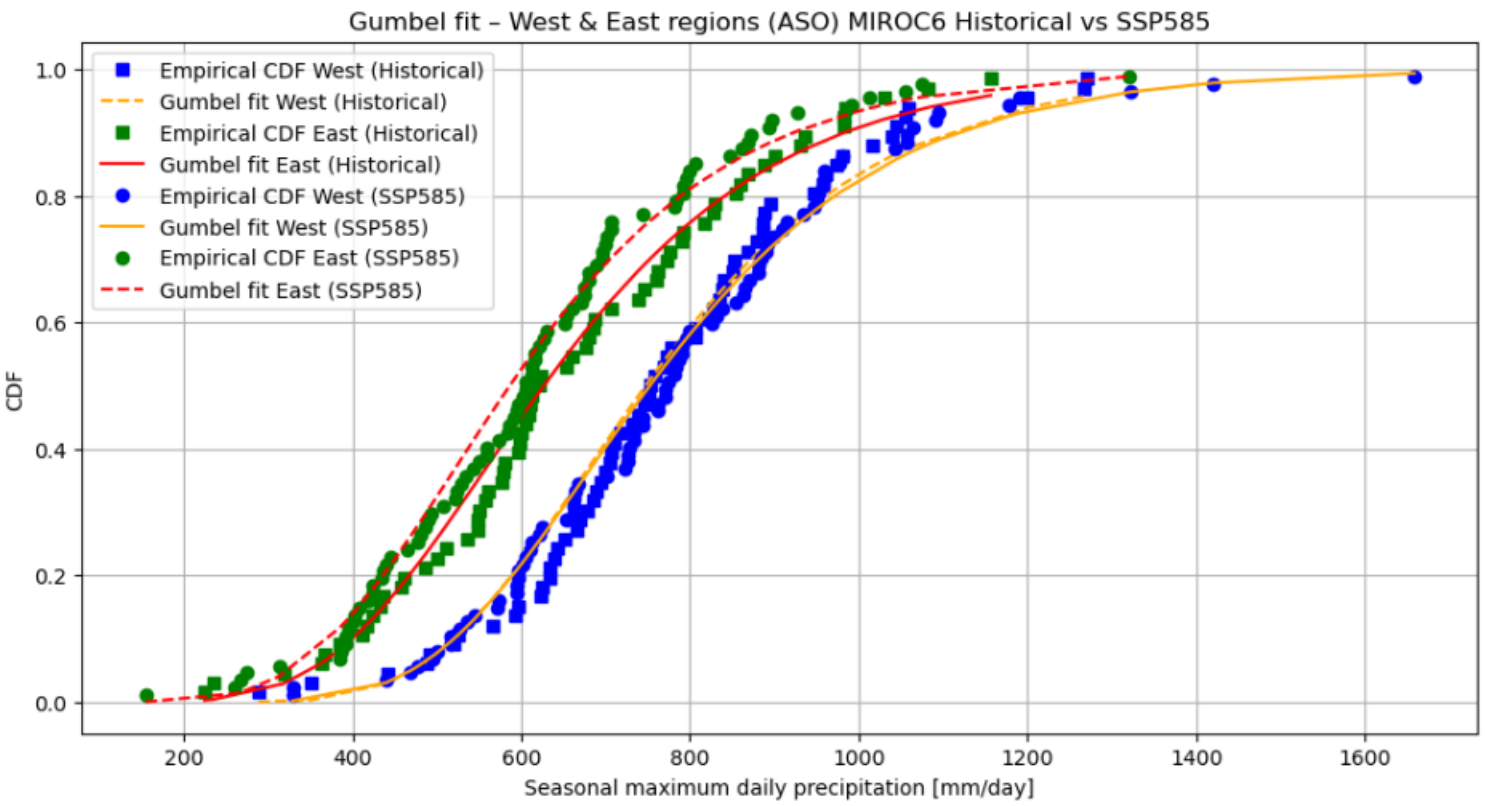
$$T = \frac{1}{1 - F(x)} \qquad F(x) = 1 - \frac{1}{T}$$

Gumbel and return periods

MIRO C6

	Return period (years)	West (mm/day)	East (mm/day)
0	10	1102.732342	983.363457
1	20	1238.481059	1119.644036
2	50	1414.193850	1296.045265
3	100	1545.865859	1428.233164

	Return period (years)	West (mm/day)	East (mm/day)
0	10	1117.308935	923.142005
1	20	1256.675115	1051.771533
2	50	1437.070340	1218.269263
3	100	1572.251175	1343.035881



Return level of total accumulated precipitation over region West and East

CDF = Cumulative Distribution Function